

(1) (a) 1. Push

"git push" command is used to push the file from the local repository to remote repository.

Syntax git push <local file> <URL of remote repo>

Ex git push <sample.py> <www.....git>

2. Merge

"git merge" command is used to merge two files in local repository.

Syntax git merge <file local file 1> <local file 2>

Ex git merge <file1.py> <file2.py>

3. Commit

"git commit" is used to save the file after we make any changes in the local repository.

Syntax git commit

Ex git commit

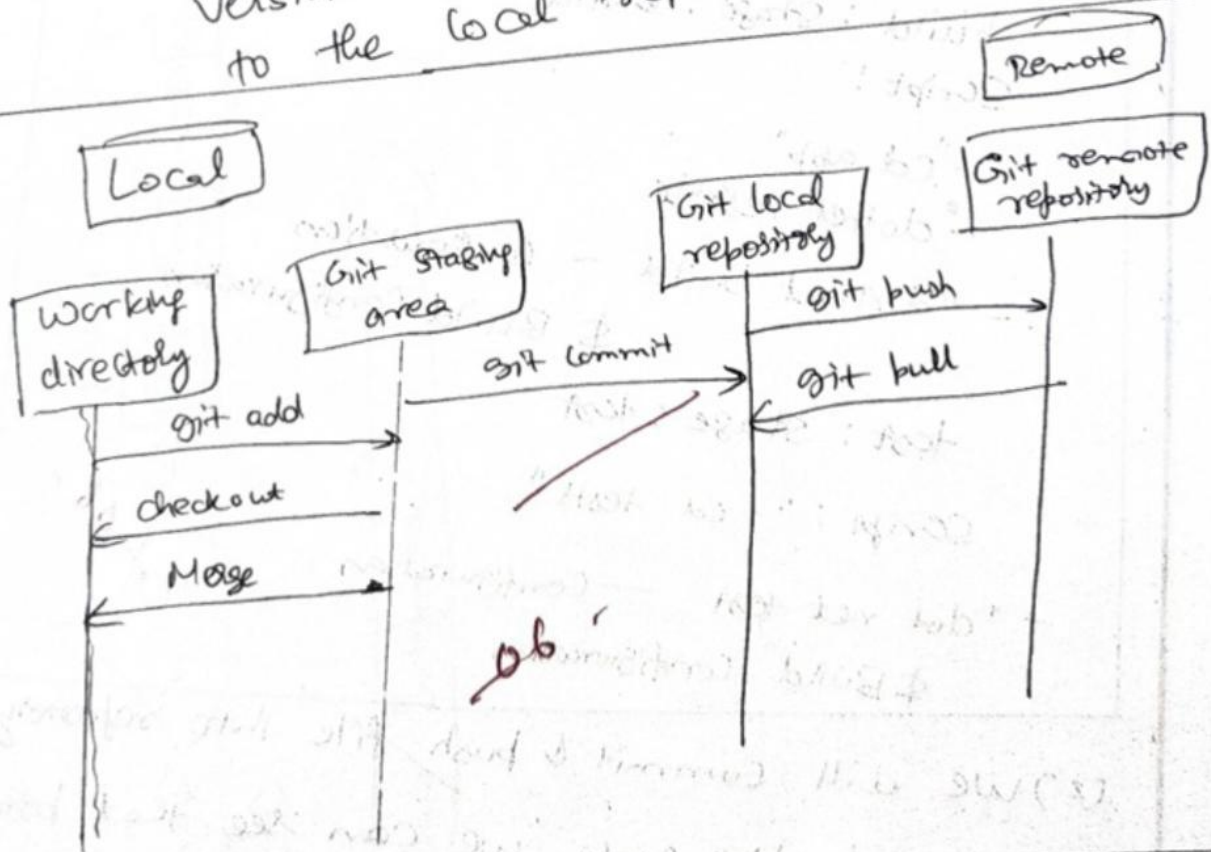
4. Checkout

"git checkout" is used to check the data of the file that we have uploaded in the local repository.

Syntax git checkout <local file>

Ex git checkout <file.py>

- (1) (b) Steps involved in understanding the Git process and Gitflow is as follows:
- (1) First developer commits the code to the local repository and pushes it to remote repository.
 - (2) Then second developer gets pushed code from remote repository.
 - (3) Second developer updates the code, creates commit & pushes new version of code into remote repository.
 - (4) Finally, the first developer develops the last version of code which is and adds it to the local repository.



(2) (a) Process of creating Continuous Integration pipeline in Gitlab

(1) To create a pipeline, we will create a file.

Content:

Image: microsoft dotnet:latest

Stages:

- build
- test

Variables: Build configuration: "Release"

build: stage: build

Script:

- "cd app"
- "dotnet restore"
- "dotnet build - Configuration
\$ Build Configuration"

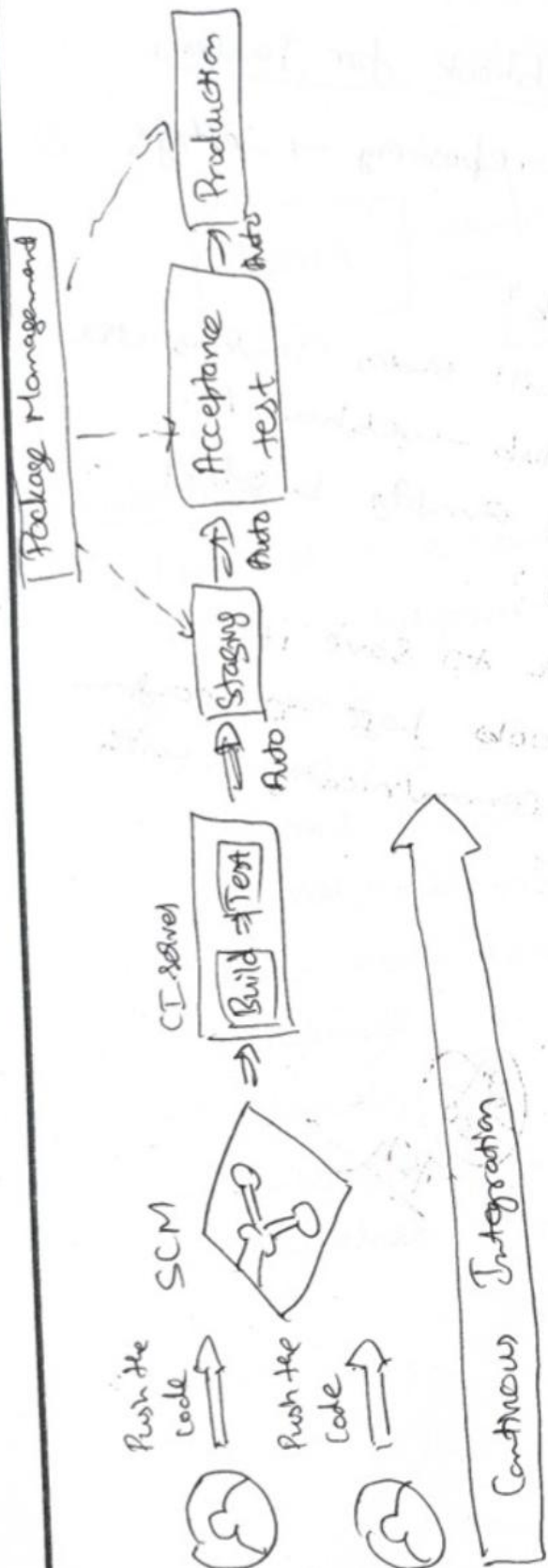
test: stage: test

Script: "- cd tests"

- "dot net test --Configuration
\$ Build Configuration"

(2) We will commit & push file into repository

(3) After pushing code, we can see that pipeline has been triggered.



② (b) Configuring Github Webhook for Jenkins

- (1) Open your Github repository → Settings
→ Webhooks
- (2) Click "Add Webhook"
- (3) In payload we will enter Jenkins URL followed by /github-webhook/
- (4) Leave secret field empty & select "Just Push event"
- (5) Click Add webhook to save it.
- (6) Go back to webhooks page to confirm it is active and communicating with the Jenkins.

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